

Project EYE

VLEO Remote Sensing Configuration Presentation

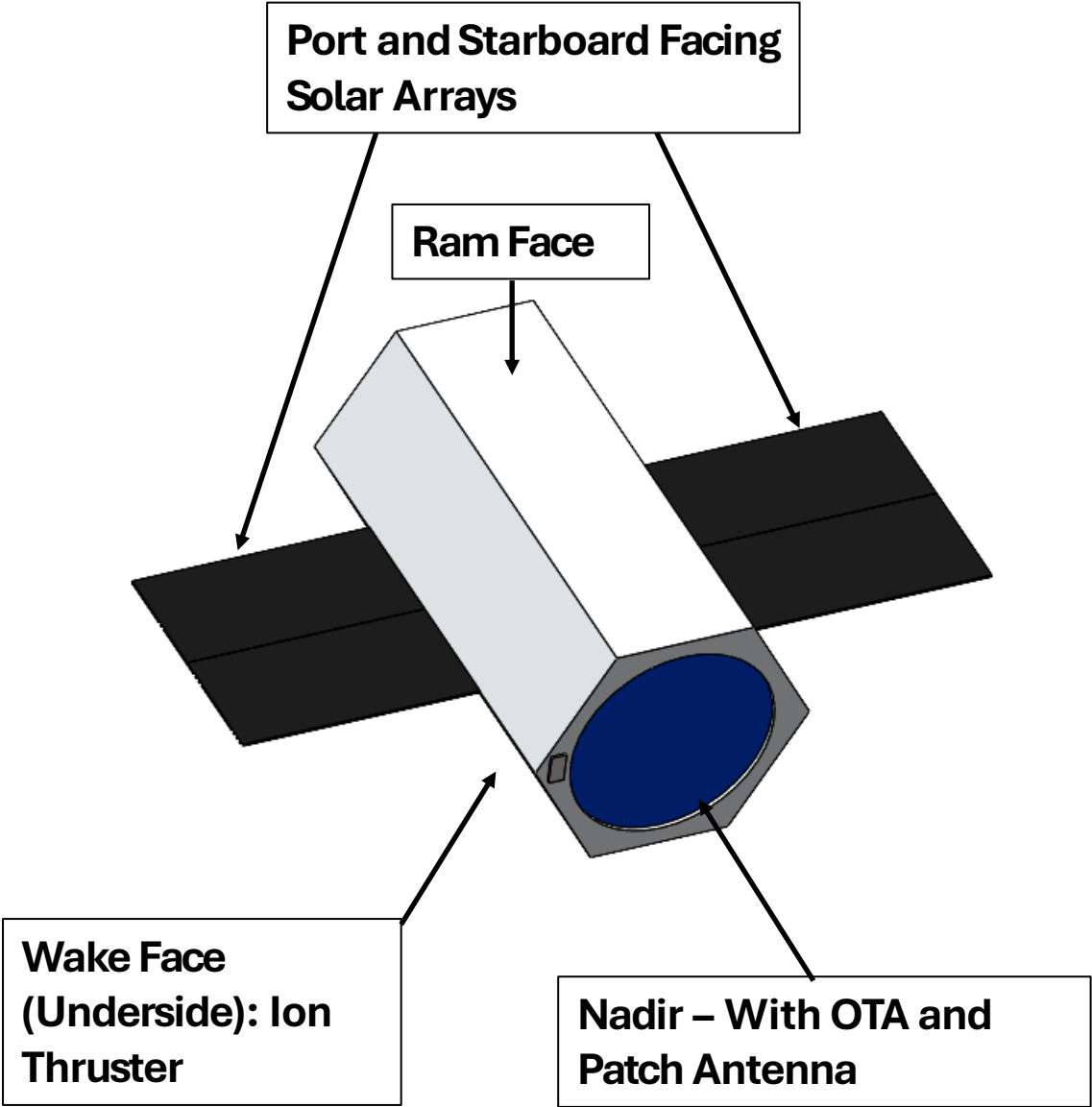
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AERO42112 Advanced Space Systems Engineering

Concept Overview

Parameter	Value
Operating Altitude	330.5km
Inclination	96.8 degrees
Total Wet Mass	99.8kg
Payload	1m GSD Optical Telescope – Utilising 8-Stage TDI Sensor
Geometry	Hexagonal Chassis
Propulsion	Ion Thrusters
Power System	Deployed Solar Arrays



Why VLEO?

Very Low Earth Orbits allow higher-resolution imagery, minimal mass and currently minimal risk of collision.
(Space Ambition and Marat Airapetian, 2024)

$$GSD \propto \frac{h}{D}$$

As the altitude of the satellite increases, the Diameter of the aperture scales proportionally for the same resolution.

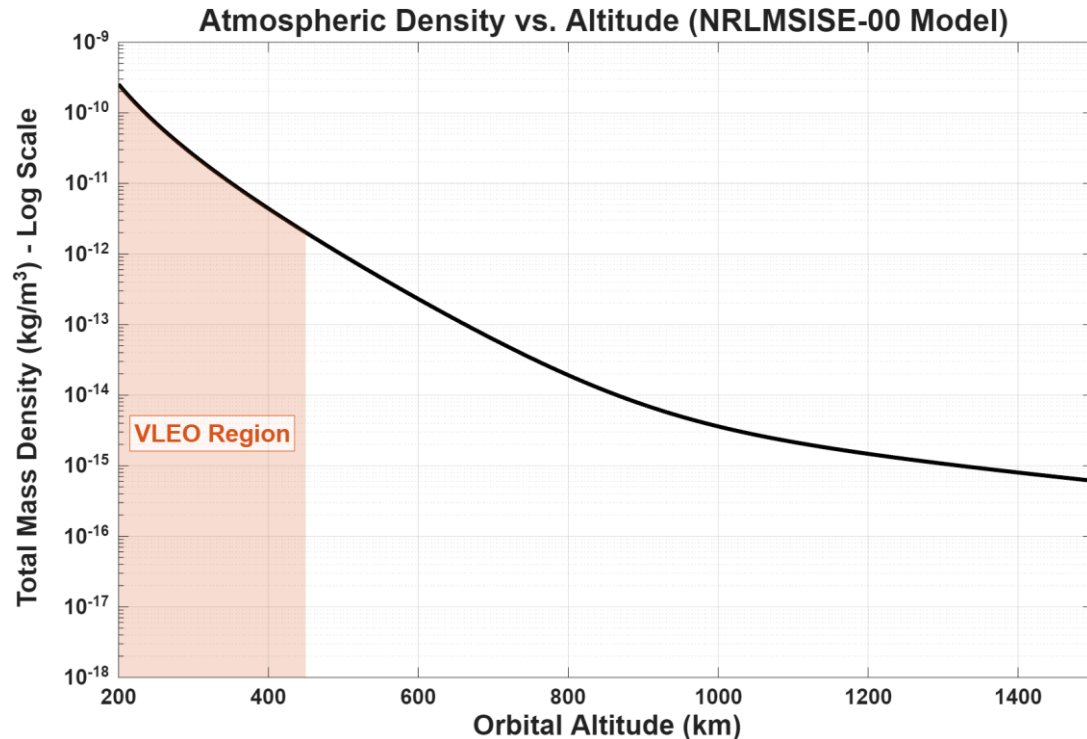


Figure 2: MSISE Density Variation

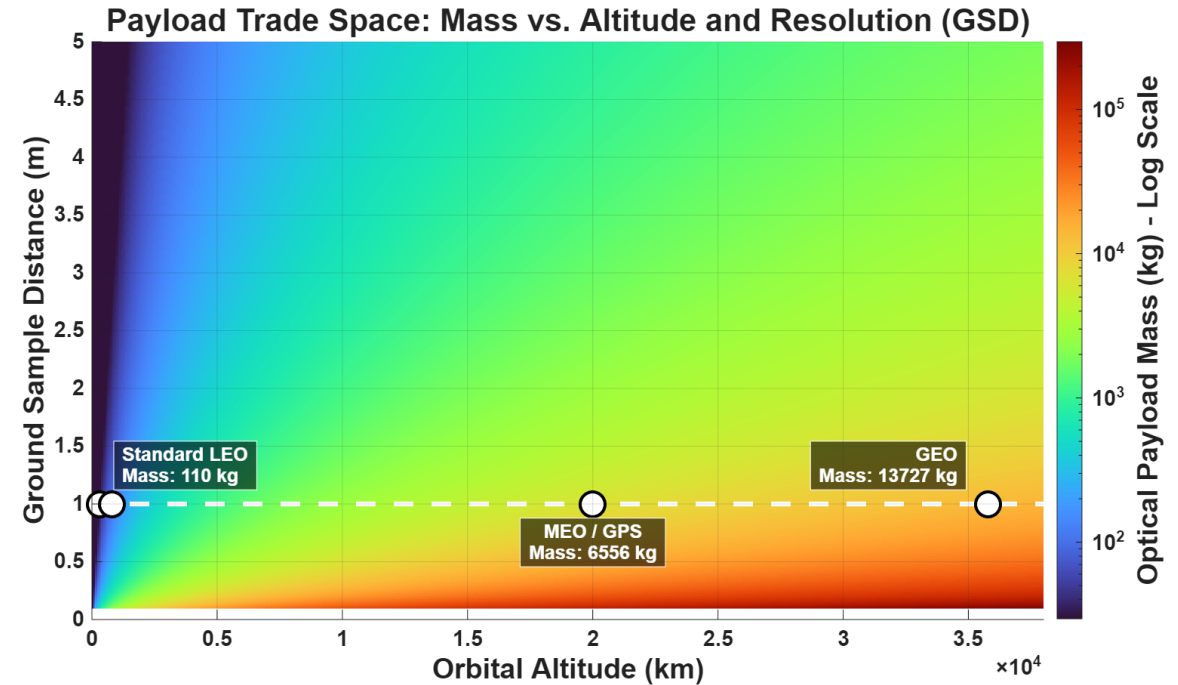


Figure 1: Mass vs Altitude Analysis

Challenges of VLEO:

- Higher Atmospheric Density + Variation
- Free Molecular Flow
- Atomic Oxygen

Orbital Design

SR2 ensures we must have an SSO orbit with a repeating ground track within 5 days.

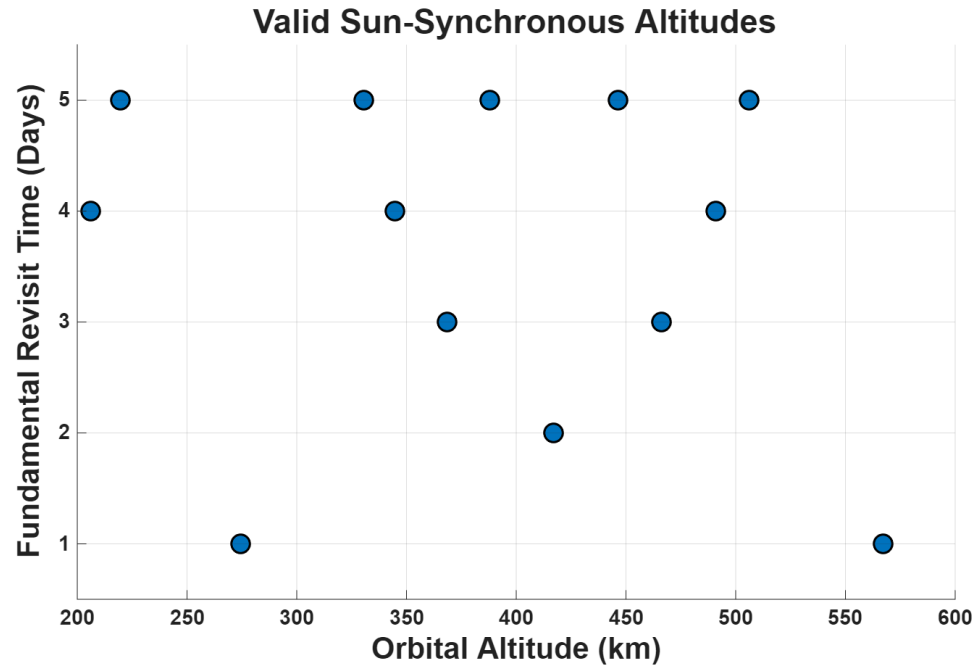
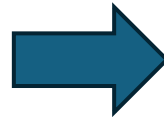


Figure 3: Repeating SSO Orbits



SR2 is verified by checking coverage. Filtering the altitude range for complete coverage.

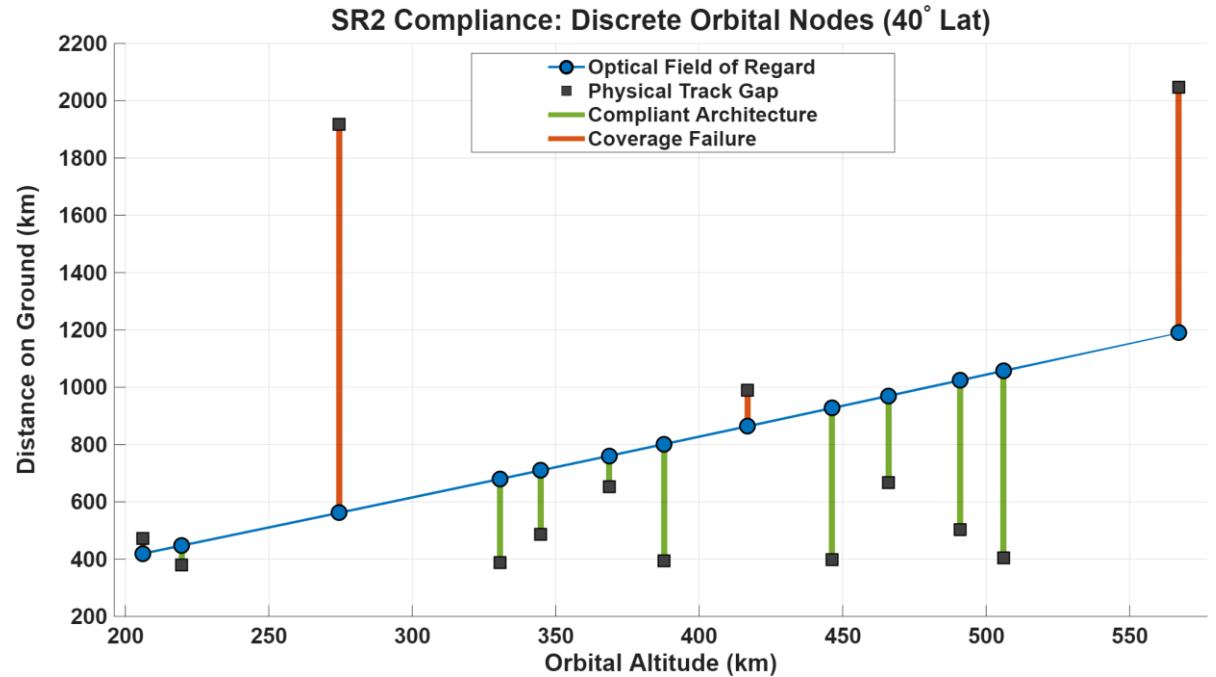


Figure 4: 40-to-80-degree latitude verification

- Altitudes in the range 200km-600km were filtered to repeating Ground Track SSO Orbits.
- Repeating Ground Track SSO Orbits were checked against FoR capability with a 45-degree off-nadir view.
- These altitudes are passed forward.

Optical Telescope Assembly Design – Two Options

Payload design is enforced by two metrics:

- Ensuring a 1m Ground Sampling Distance – SR1
- Ensuring Signal-To-Noise Ratio is greater than 100 – SR4

Assumed Payload Length is equal to twice the Diameter.

Enlarged Aperture

Increases ability for light to be captured

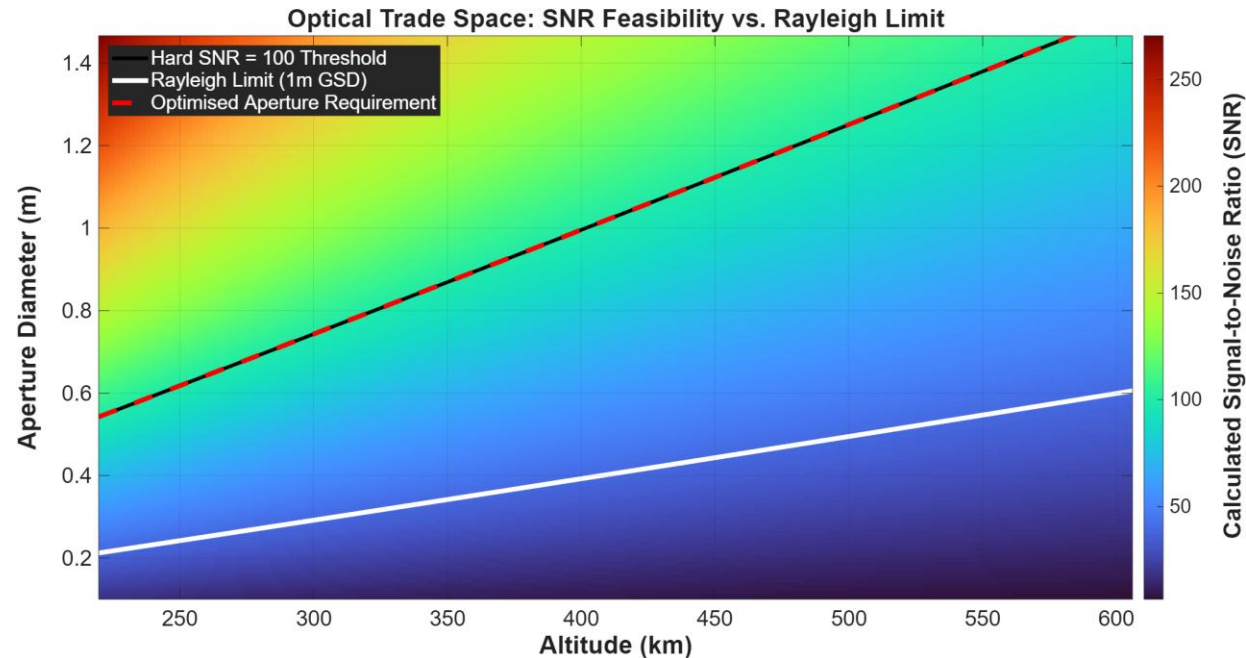


Figure 5: Enlarged Aperture Heatmap

Limited by SNR Requirement

Time Delay Integration

Effectively increases Integration time

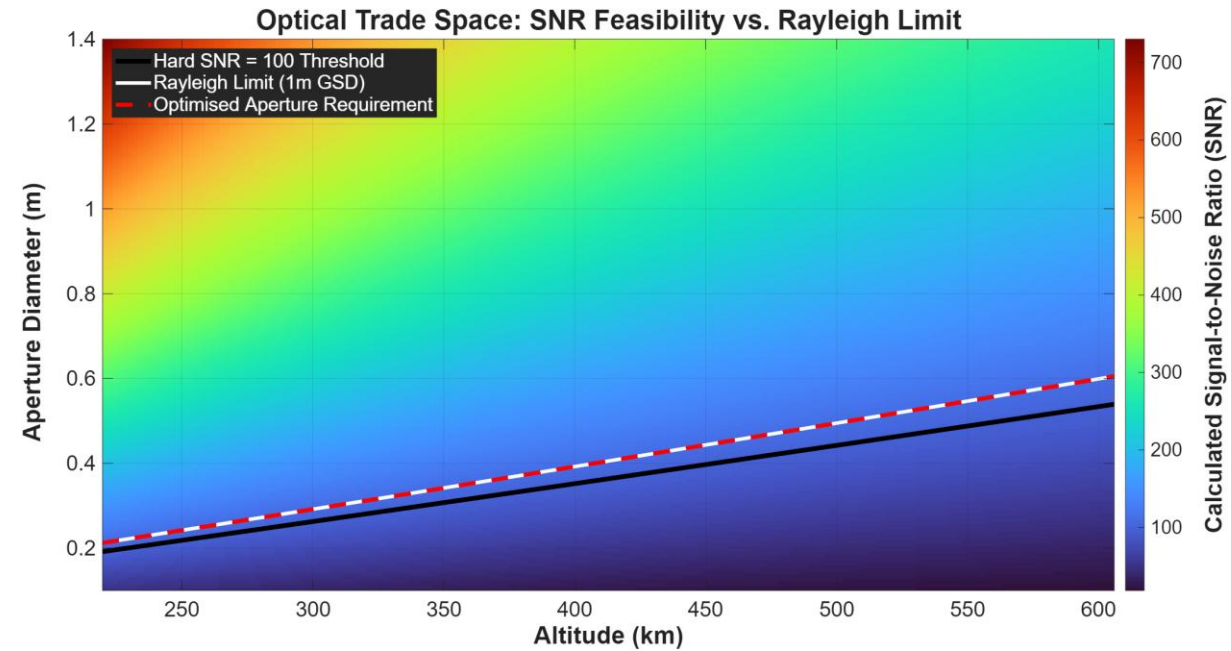


Figure 6: TDI Heatmap

Limited by Diffraction Limit (Rayleigh Criterion)

Aperture Mass Breakdown

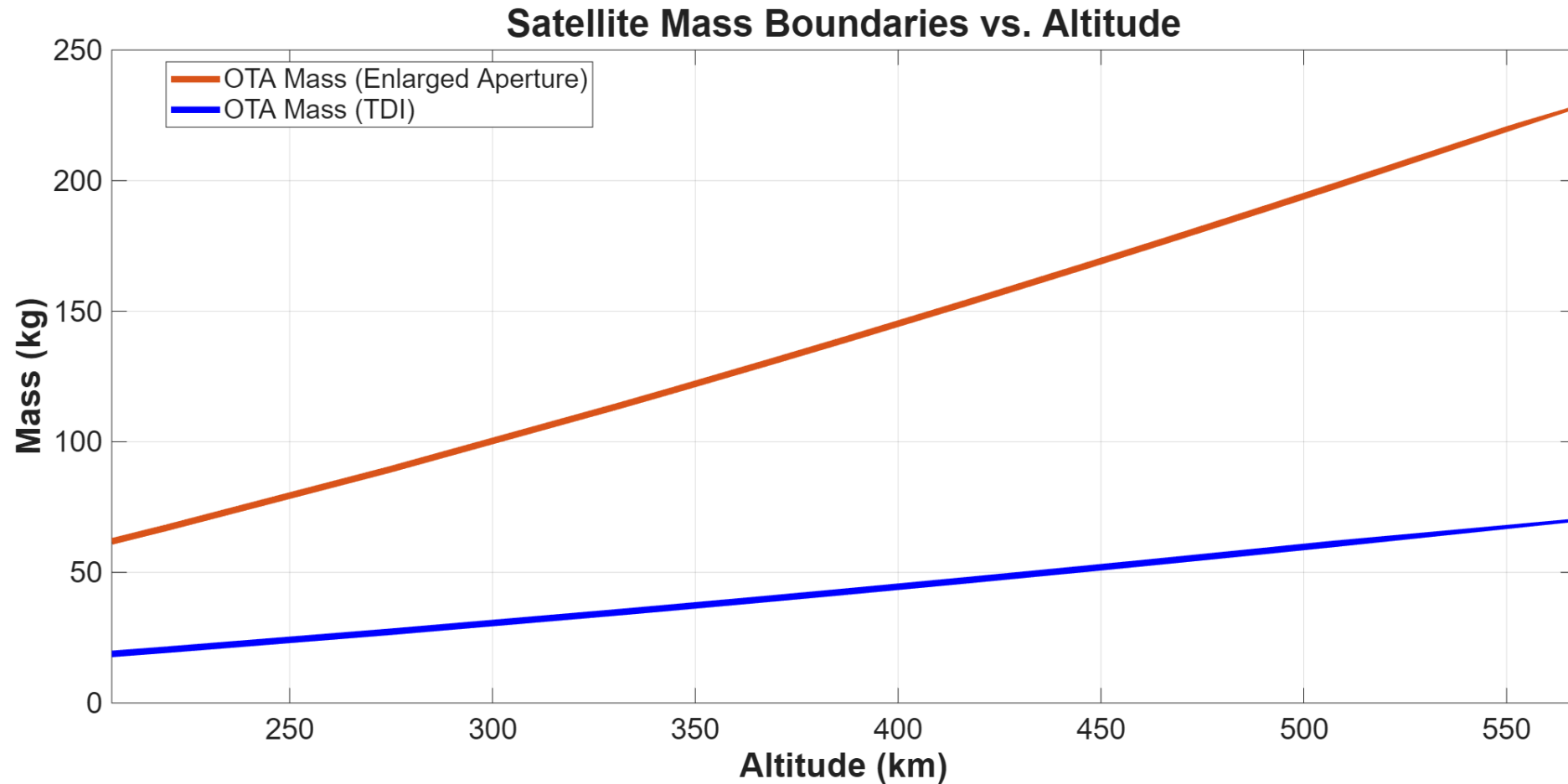


Figure 7: OTA Mass Trade

Due to the relation: $OTA\ Mass = 146 * D^{1.27}$
Time Delayed Integration causes the mass of the OTA to be heavily reduced

Aerodynamic Modelling

Free Molecular Flow Analysis (Panel Method) was undertaken using SOLIDWORKS models and ADBSat.

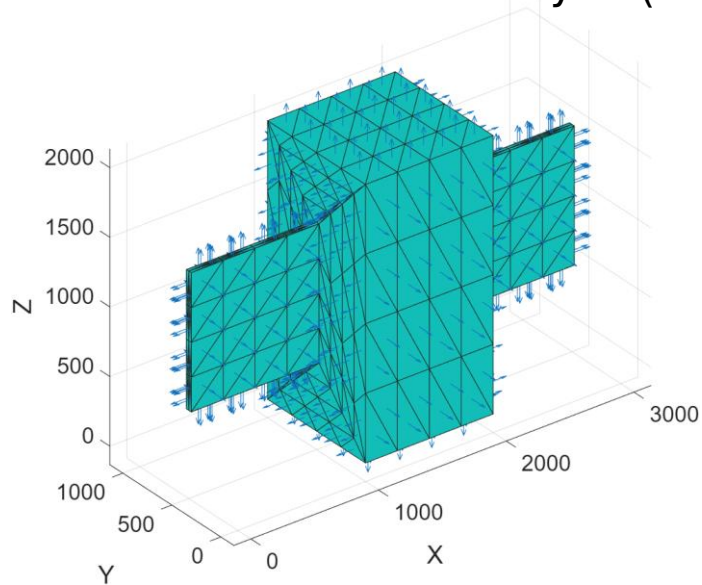


Figure 8: ADBSat Mesh Example

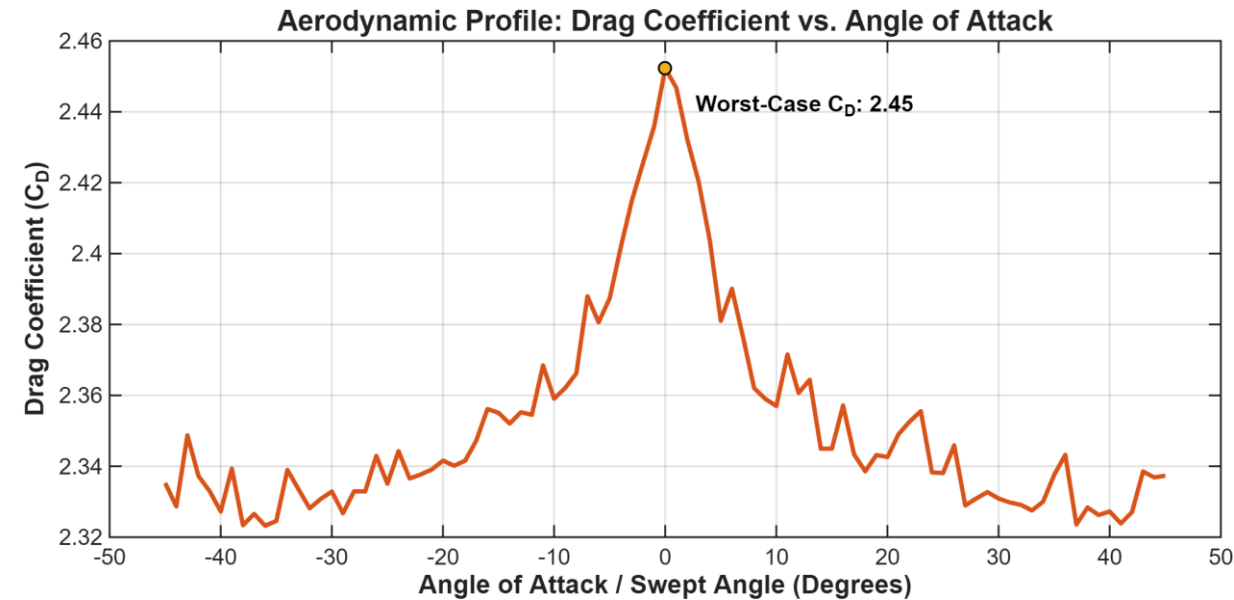
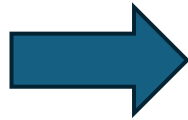


Figure 9: C_D for Cuboid with Deployed Arrays

Model	Coefficient of Drag Value
Cuboid with Deployed Arrays	2.45
Cuboid with Body-Mounted Arrays	2.38
Cylinder with Deployed Arrays	2.28
Cylinder with Body-Mounted Arrays	2.2
Hexagon with Deployed Arrays	2.36

Table 2: C_D of different Configurations

Simulation Parameters:

- Altitude: 400km
- Inclination: 97 degrees
- Sizing: Bus Geometry based on 400km aperture requirements

Power and Propulsion

Across the valid altitude range, for all geometric configurations, a trade-off occurs between the five chosen thruster types.

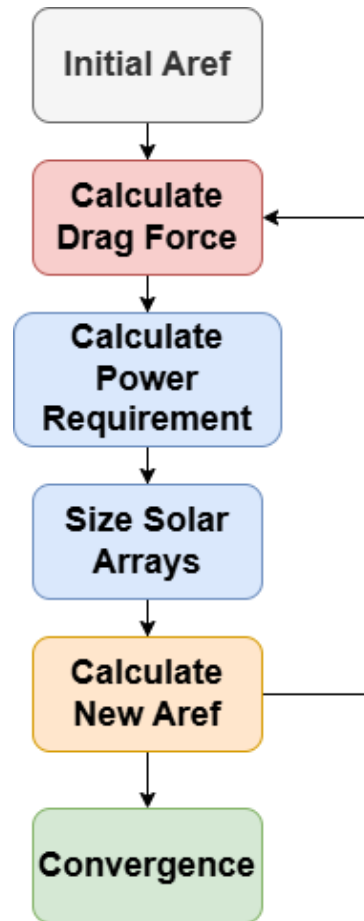
Propulsion	I_{sp}	Efficiency
ResistoJet	300	0.775
Arc Jet	550	0.35
Ion Thruster	3050	0.6
Hall Thruster	1750	0.475
PPT	1075	0.1

Table 3: Different Propulsion Types

Current Industry trends are moving to include Air-Breathing Electric Propulsion (ABEP). This utilises the atmosphere as propellant reducing the drag penalty of VLEO.

However, due to it currently being of a low Technology Readiness Level it is being disregarded.

The Drag Loop



- Enforces 20% volume margin
- Enforces Stowage requirement

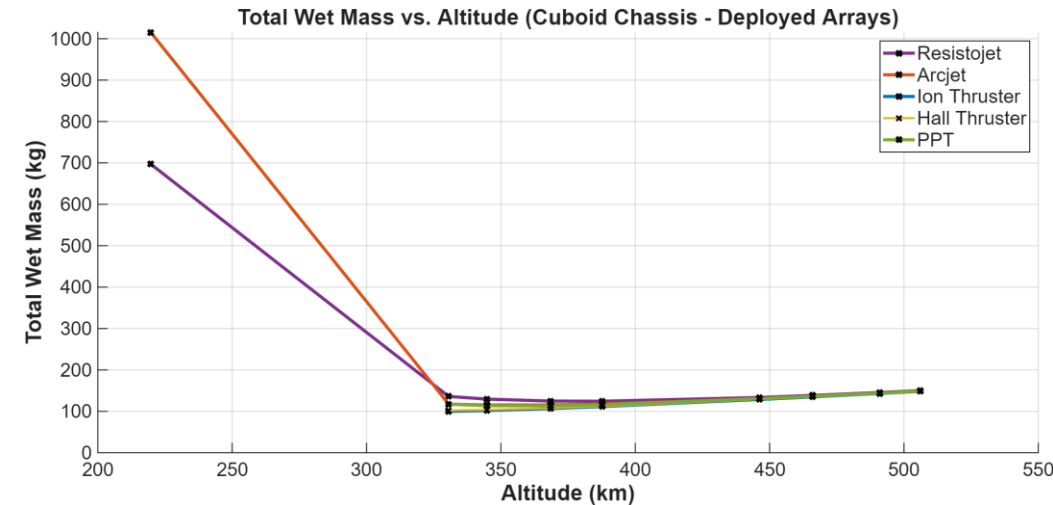


Figure 10: Wet Mass vs Altitude For Cuboid Configuration

For all configurations it was shown that 330.5km altitude was optimal for finding balance between drag and power requirements, minimising wet mass for a 5 year mission – SR5.

Selected Altitude: 330.5km (5-day revisit)

Selected Thruster Type: Ion Thruster

Solar Array Configuration – Deployed vs Body

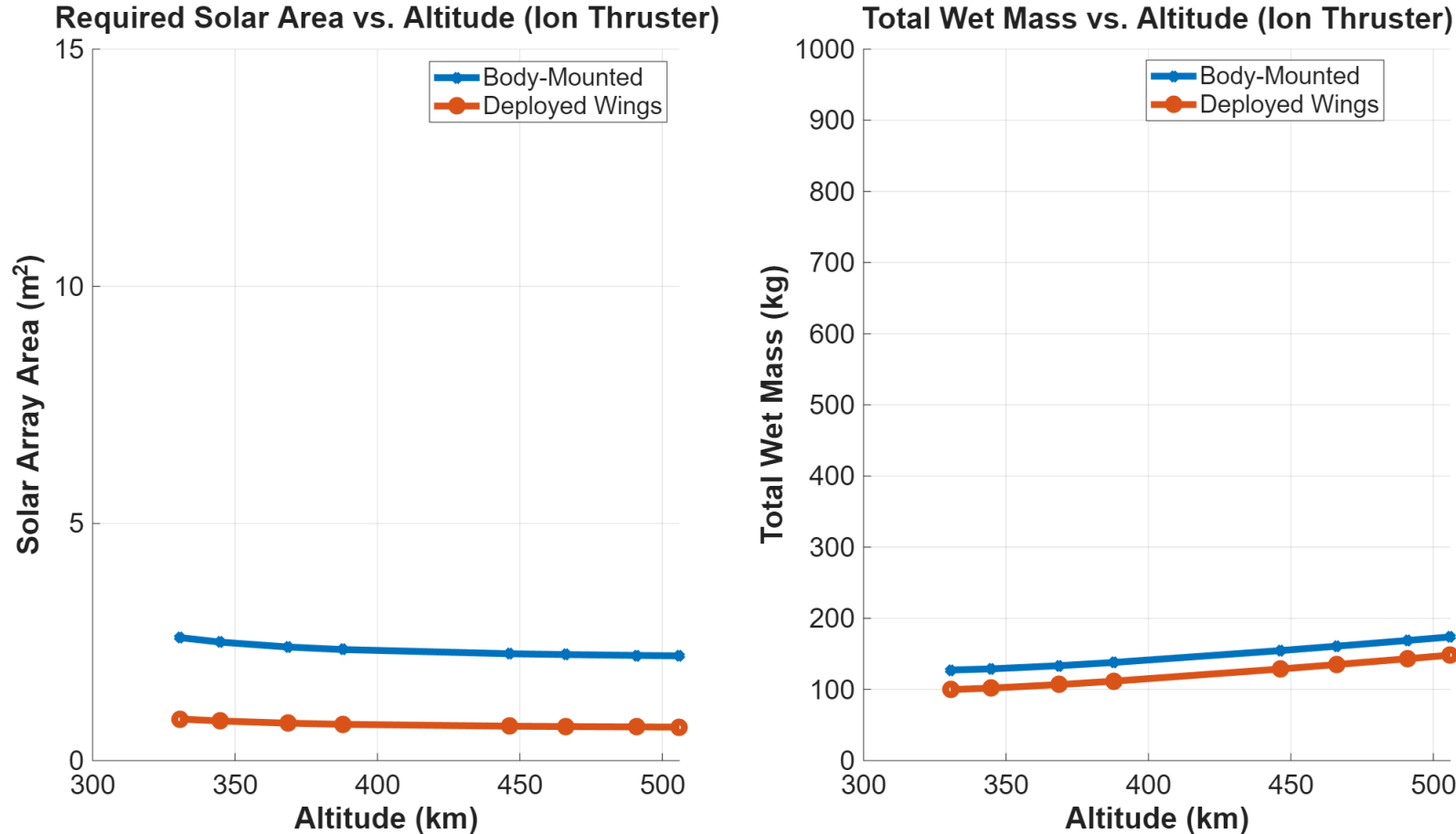


Figure 11: Comparison between Solar Array Configuration

When choosing between deployed body mounted, a geometric penalty is applied to the body mounted arrays due to the inability for sun tracking or complete illumination.

Similarly Deployed Arrays have the penalty of their area is summed with the bus area when calculating A_{ref} in the drag cycle.

Selected Configuration: Deployed

Systems Model Integration

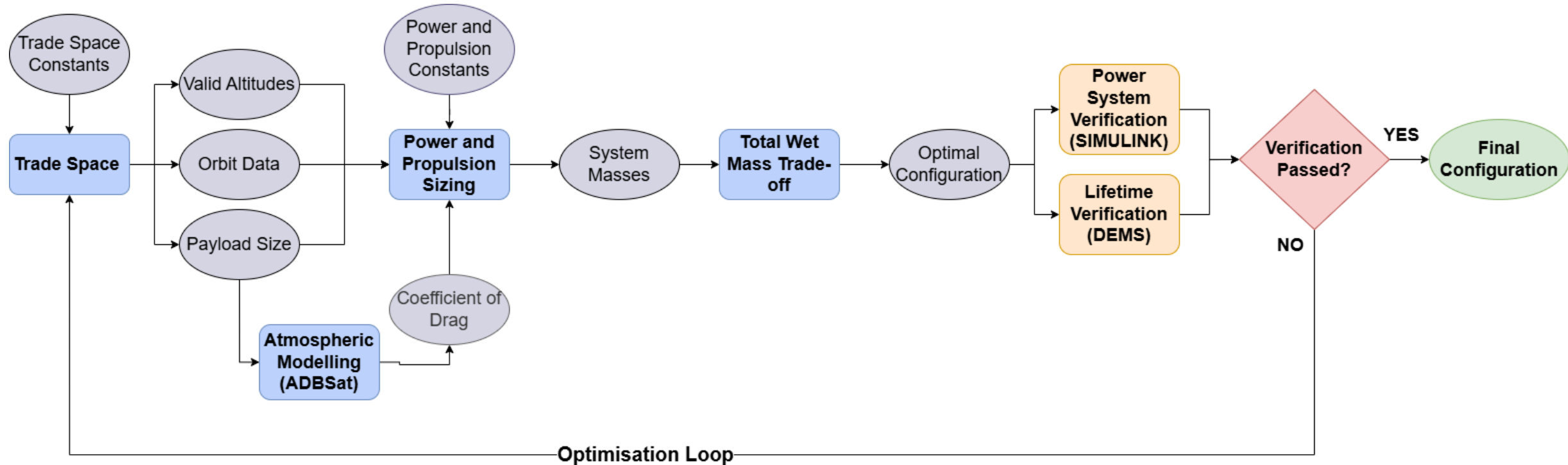


Figure 12: Complete Systems Model

Trade Space

- Based on Initial conditions
- Enforces SR1-SR4 (Resolution, Revisit, Mass, Orbit Constraints)

Power and Propulsion Sizing

- Based on Power assumptions
- Utilises Trade Space Outputs
- Automated convergence within 0.001 tolerance threshold.
- Begins enforcing SR5

Verification

- Enforces SR5
- Verifies against Atmospheric variation (Day/Night)
- Verifies Orbit lifetime
- Ensures reasonable configuration

Critical Assumptions

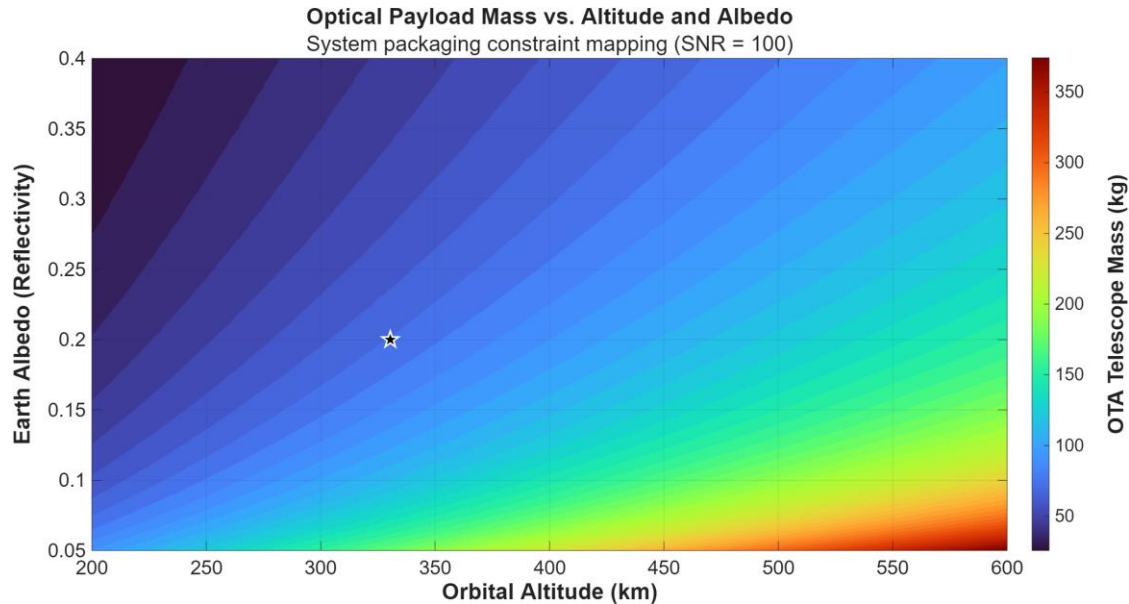


Figure 13: Albedo Heatmap

Albedo Scales with required Diameter, nominal value of 0.2 was chosen based on WorldView.

Other Assumptions:

- C_D calculated based on: $A_{ref} = A_{ref_{bus}} + A_{ref_{solar}}$
- Payload is only active during the lighting phase (60W Power).
- ADCS can maintain required pointing accuracy for TDI.
- The Nadir face requires $0.01m^2$ minimum area for patch antenna.
- 10cm minimum clearance around aperture for vibrations.

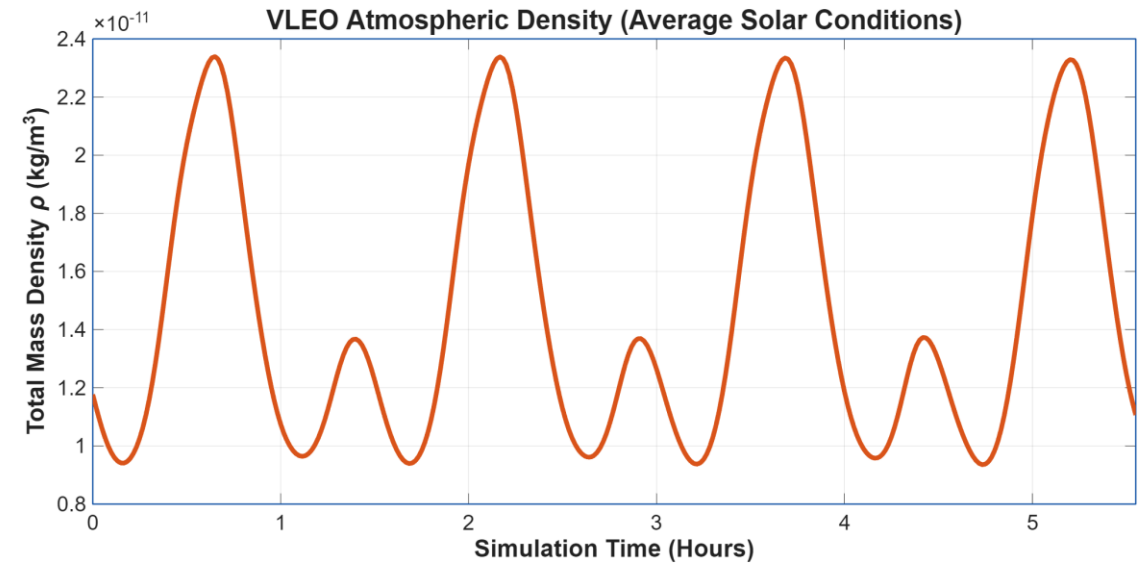


Figure 14: Nominal Atmosphere Variation

Nominal Solar Conditions based on the solar maximum of July 2025 with Day/Night variation:

- A_p of 15 – Maximum value classed as “Unsettled” (Geomagnetic effect)
- F_{107} and F_{107A} chosen as 150 Based on:
$$F_{107} = 68.84 + 0.67 * \text{Sunspot Number}$$

(Dhivya, 2017; NOAA/NWS Space Weather Prediction Center, no date)

Trade Space & Alternatives

Subsystem	Option A	Option B	Option C	Option D	Option E
Optics	Aperture Inflation	8-Stage TDI	N/A	N/A	N/A
Propulsion	ResitoJet	ArcJet	Ion Thruster	Hall Thruster	PPT
Solar Array	Deployed	Body-Mounted	N/A	N/A	N/A
Configuration	Cuboid	Cylinder	Hexagonal	N/A	N/A
Altitude	219km	330km	345km	369km	400-600km

Table 4: Trade Space

Key Budgets

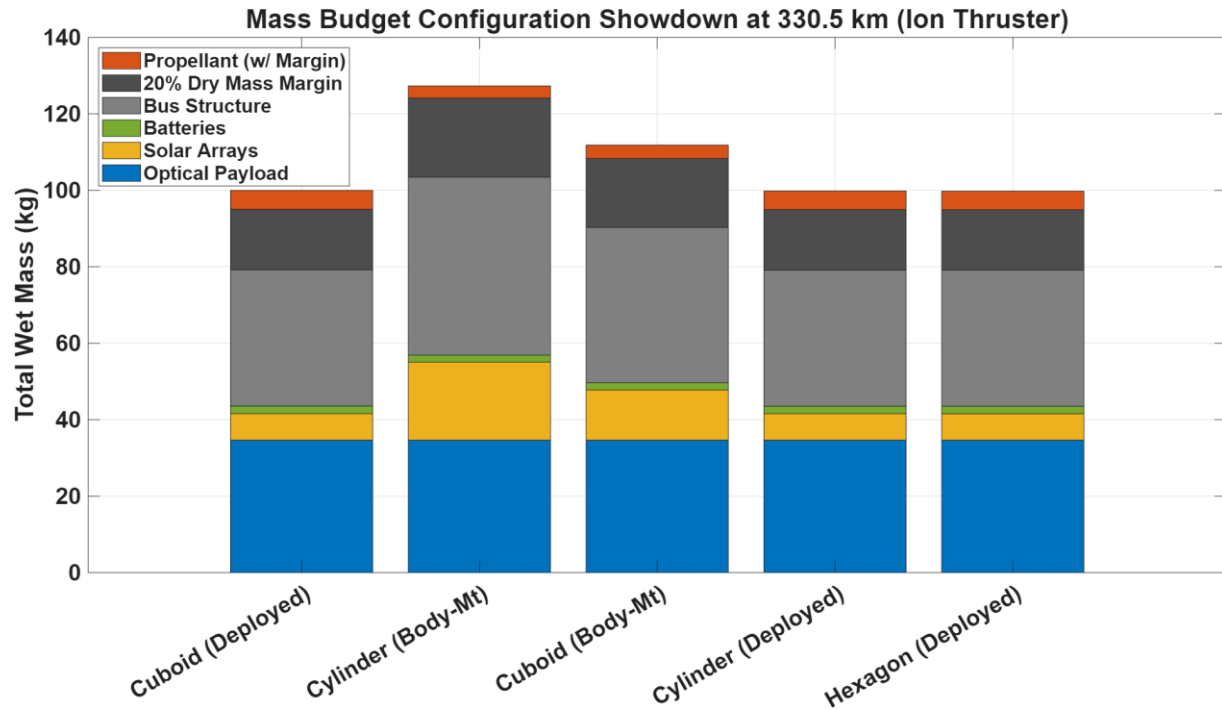


Figure 15: Wet Mass Trade-off

- Deployed Arrays clearly showcase reduced solar panel mass with increased propellant mass to counteract drag.
- Deployed configurations are very similar in mass due to A_{ref} being sufficiently small that C_D variation among them has little effect.

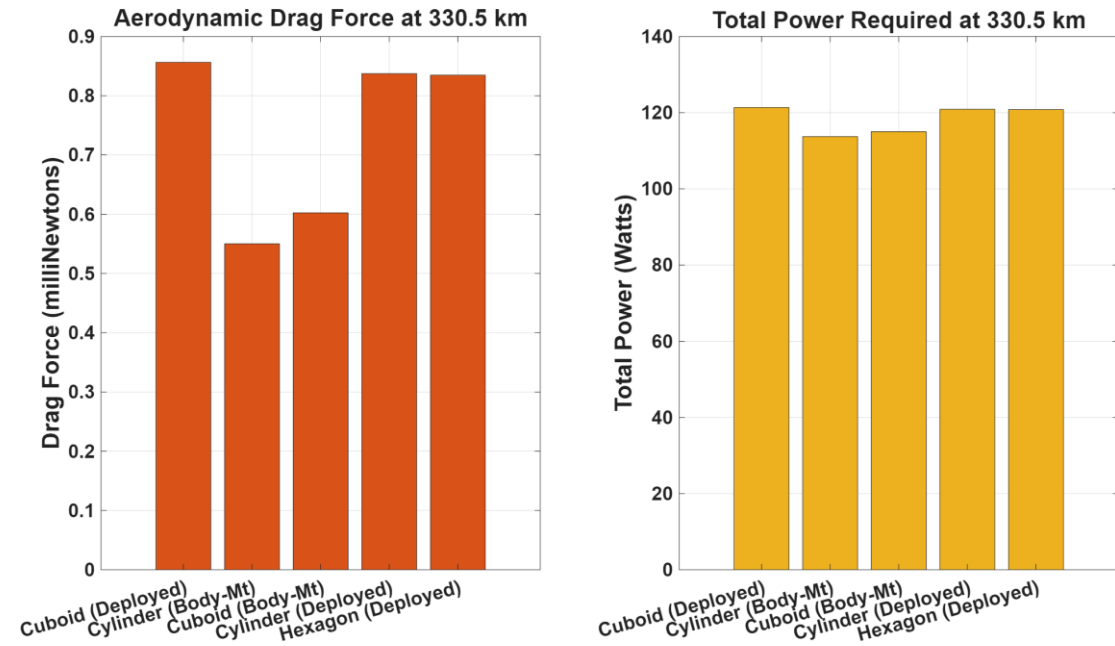


Figure 16: Power and Drag Trade-off

- Aerodynamic Drag and thrust therefore needed is much higher for Deployed Array Configurations.
- However, power required is only marginally different.

Selected Solution – Hexagonal Chassis

Dry Mass (Kg)	Dry Mass Margin (20%)	Propellant (+10% Margin)	Total Wet Mass (kg)
79.2	15.8	4.8	99.8

Table 4: Selected Configuration Mass Breakdown

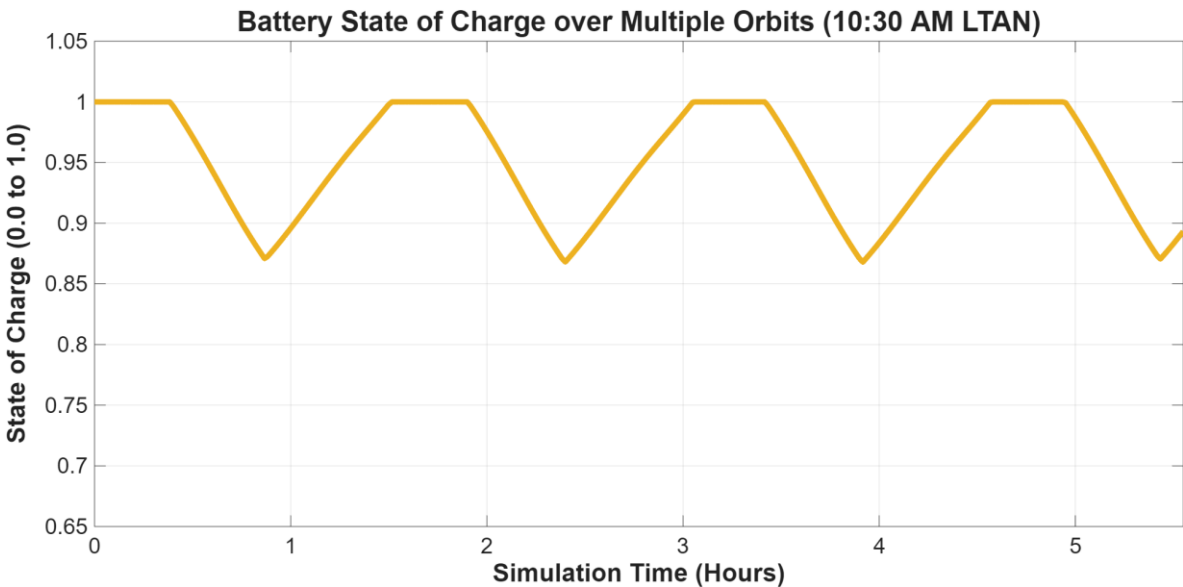


Figure 17: Power System Verification

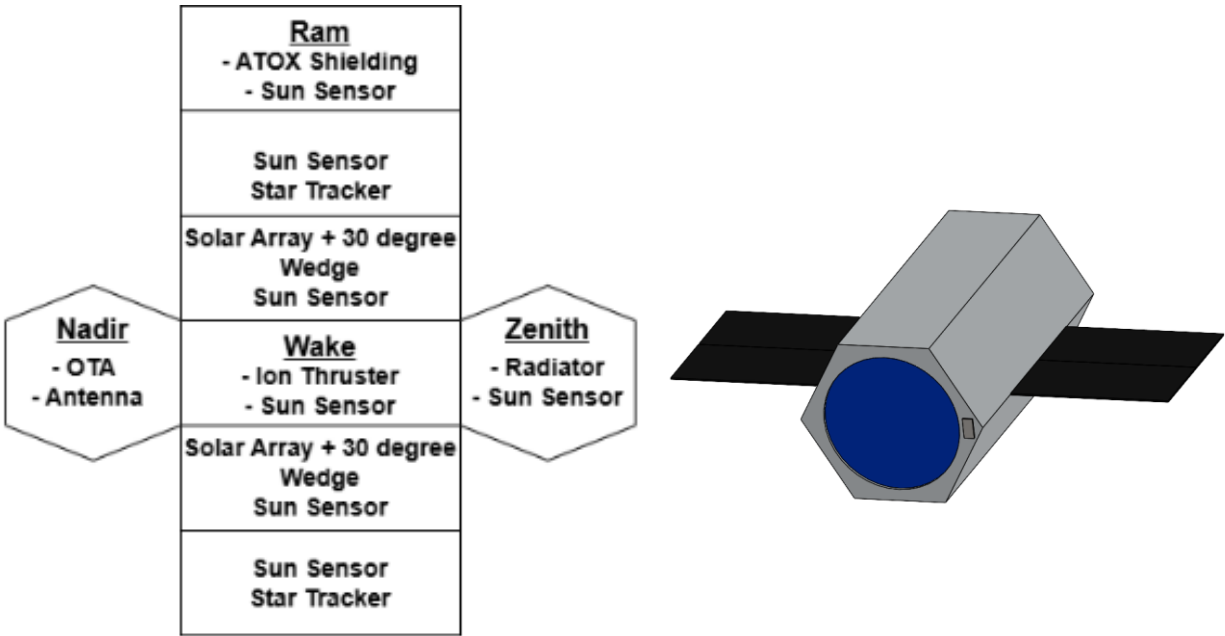


Figure 18: Component Arrangement and CAD Model

The Hexagonal Chassis allows a reduced C_D whilst still allowing flat faces for component fixture. When accounting for volume and clearance requirements its provided minimum wet mass overall.

Sustainability & End of Life

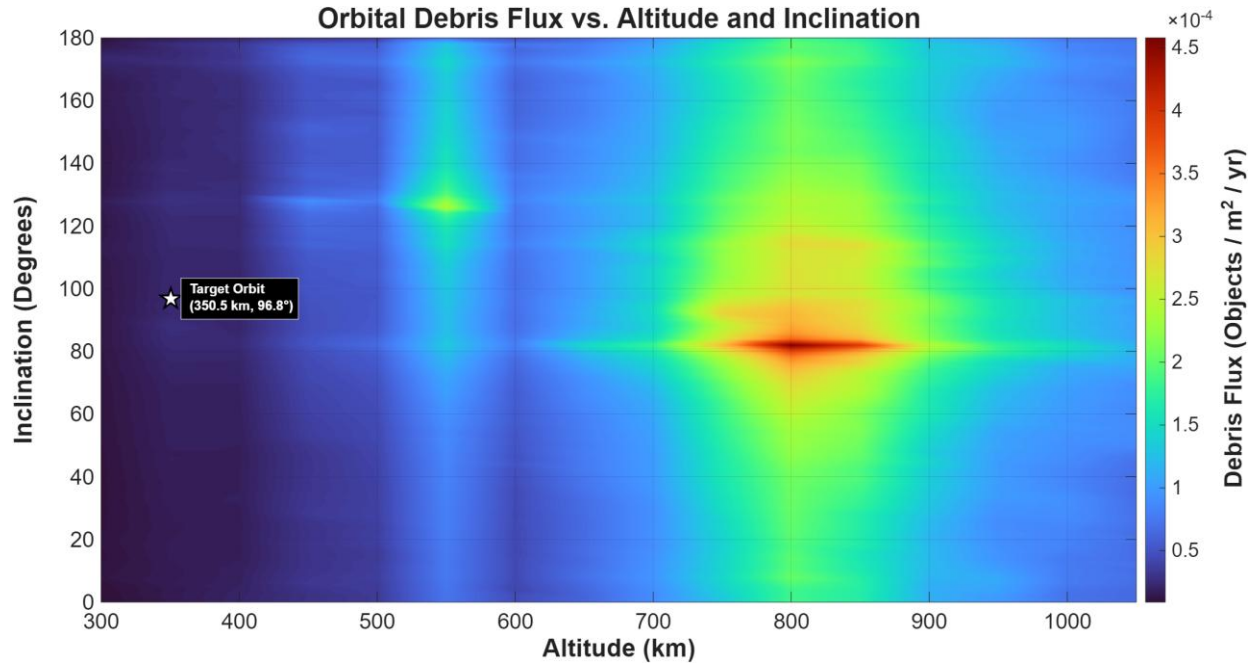


Figure 19: Density Flux Heatmap (2024)

- De-orbit in VLEO is simple due to high atmospheric drag. Complying with FCC 5-year post mission de-orbit timeline burning up in 72 days.

(Federal Communications Commission, 2022)

- Design for Demise (D4D)** also ensures that post mission, the satellite burns up during re-entry. Complying with ESA guidelines.

(European Space Agency, 2020)

- Operational Safety:** As seen in Figure 19, operation at 330.5km and inclination 96.8 degrees ensures a low debris flux. Drastically reducing collision probability and need for excess propellant for collision avoidance

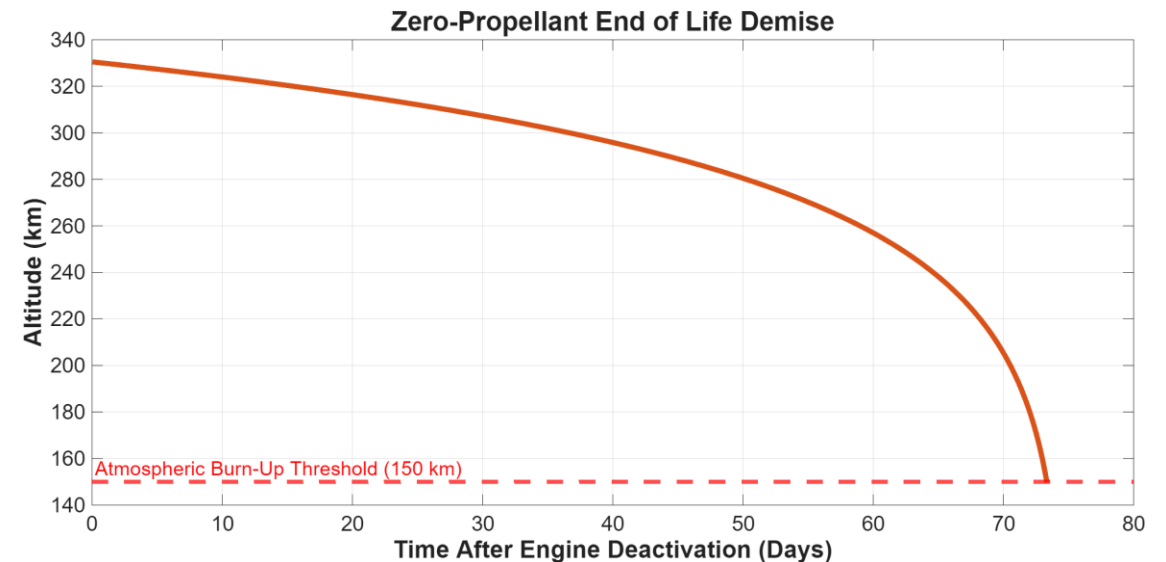


Figure 20: End of Life Verification

System Requirement Compliance

Requirement	Status	Met? [Y/N]
SR1 – 1m GSD Resolution	Achieved at 330.5km Altitude with 8-Stage TDI Optical Assembly.	Y
SR2 – 5-day Revisit Time for target latitudes	Altitudes were filtered for 5-day revisit times and 330.5km was chosen.	Y
SR3 – Repeating Ground Track of LTAN 10:30am	Chosen inclination and altitude ensures SSO repeating Ground Track. Launch vehicle ensures LTAN.	Y
SR4 – SNR shall be greater than 100	Utilising Time Delay Integration ensures SNR is over 100 for minimal mass.	Y
SR5 – 5-year lifetime	Propellant tank sized for 5-year continuous burn, along with investigated safe mode (Appendix B) and orbit verification.	Y

References

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- Space Ambition and Marat Airapetian (2024) *The New Space Frontier: Why VLEO is an Emerging Trend*, *Substack.com*. Space Ambition. Available at: <https://spaceambition.substack.com/p/the-new-space-frontier-why-vleo-is>.

Any Questions?

Appendix Contents

A – SIMULINK MODEL

B – Safe Mode

C – DEMS Data

D – Volume Study

Appendix A: - SIMULINK Model

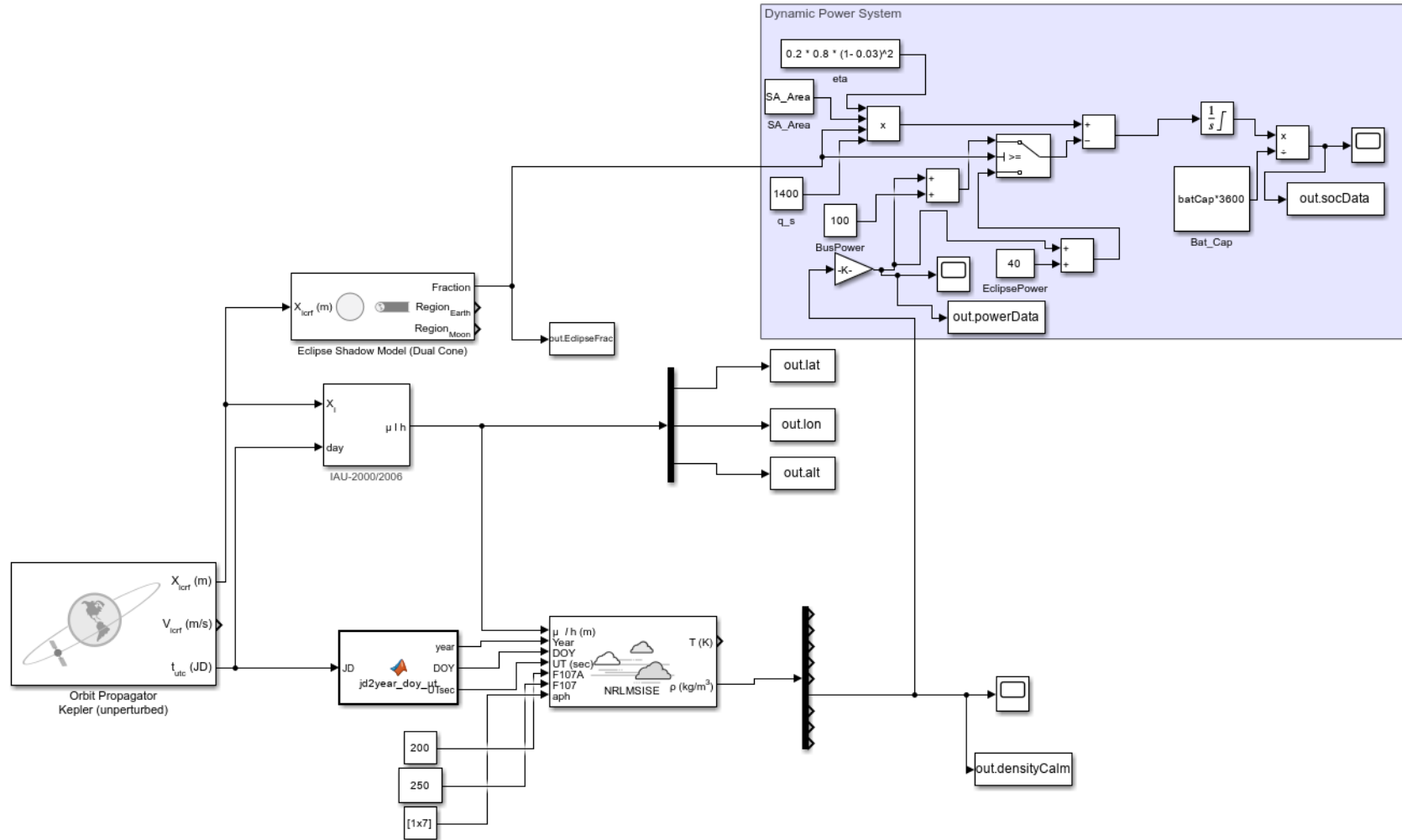


Figure A1: Complete SIMULINK Model

Appendix A- SIMULINK Model

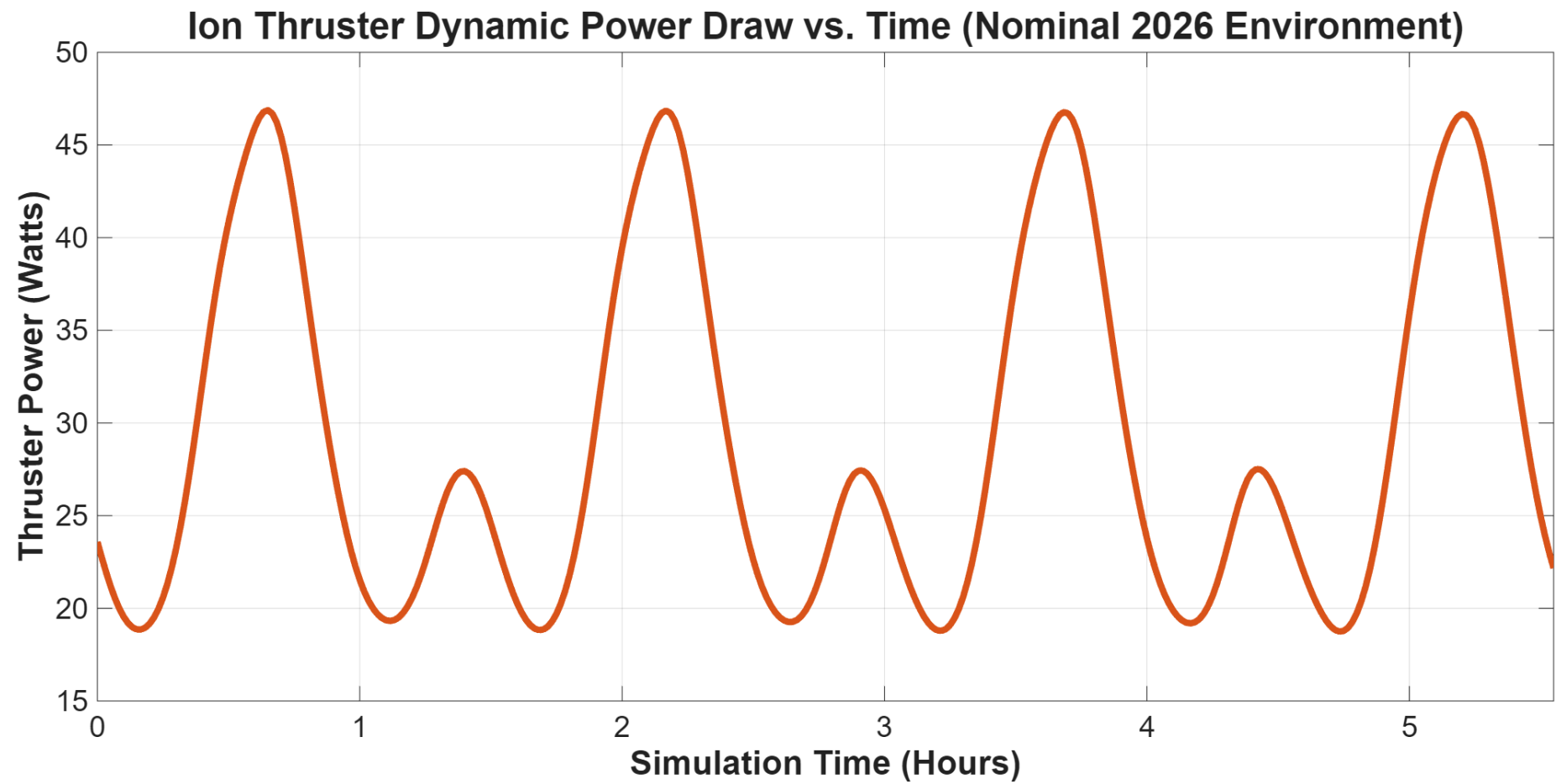


Figure A2: Dynamic Thruster Power Draw

Appendix B: - Safe Mode

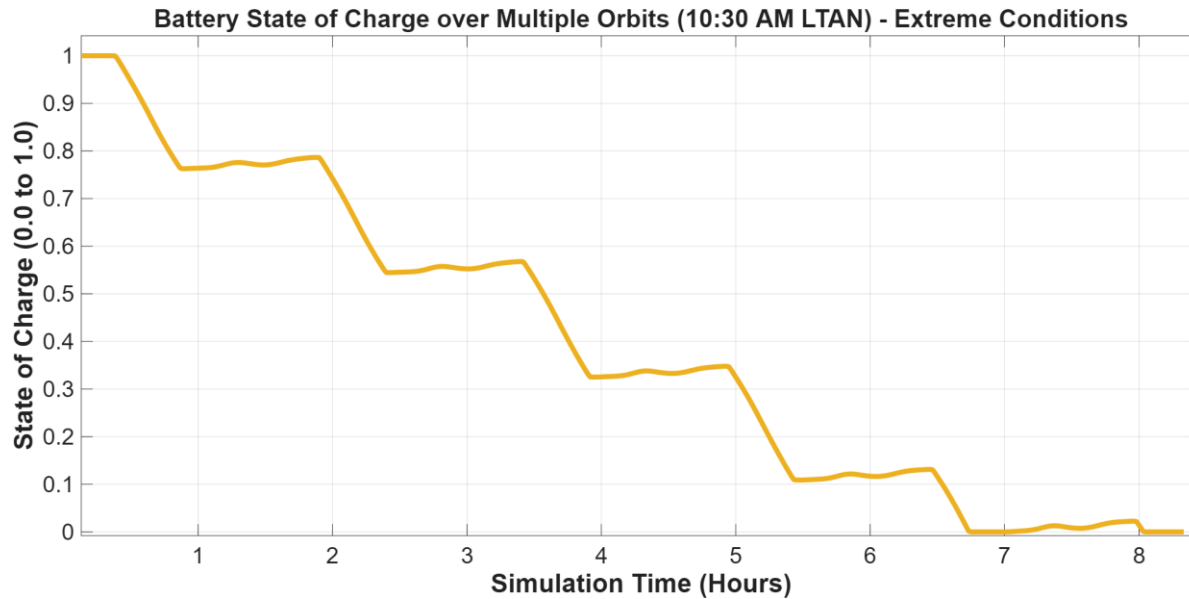


Figure B1: Battery in Extreme Conditions Without Safe Mode

An extreme event based on a Coronal Mass Ejection (CME) was simulated. This was using values of 250 for F_{107} and 300 for ap value.

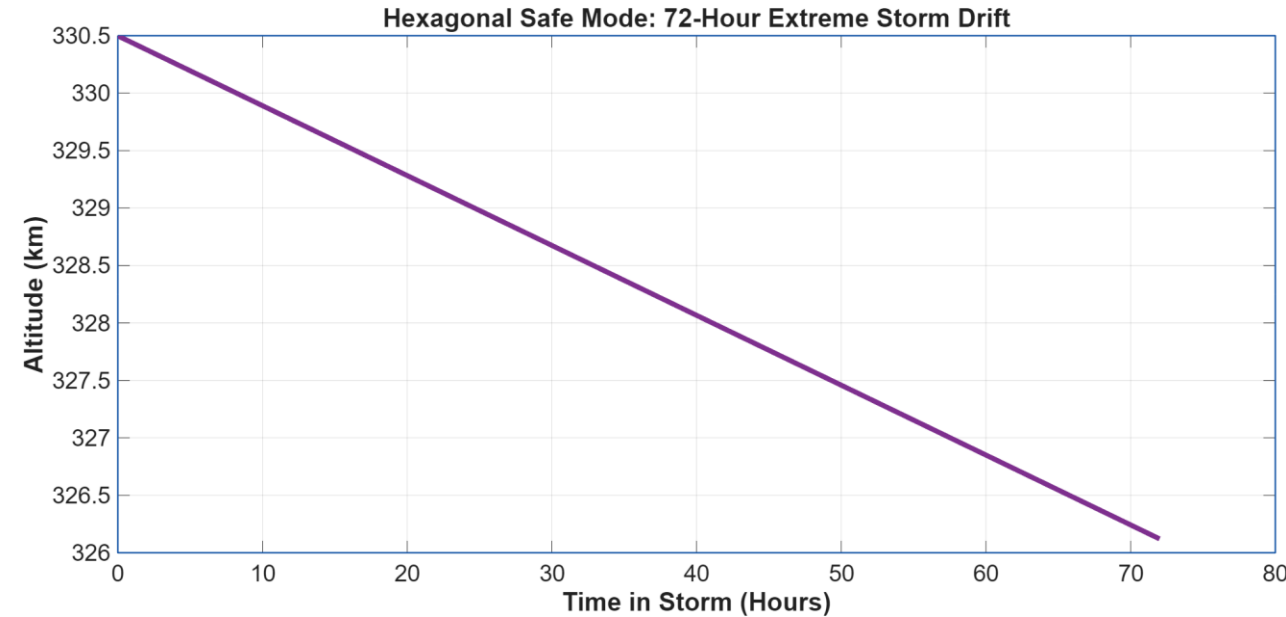


Figure B2: Safe Mode Altitude Change

Safe mode turns off all power systems apart from the main ADCS system. After a nominal storm time of 72 hours based on historical data, the satellite can return to normal operation.

Appendix B: - Safe Mode

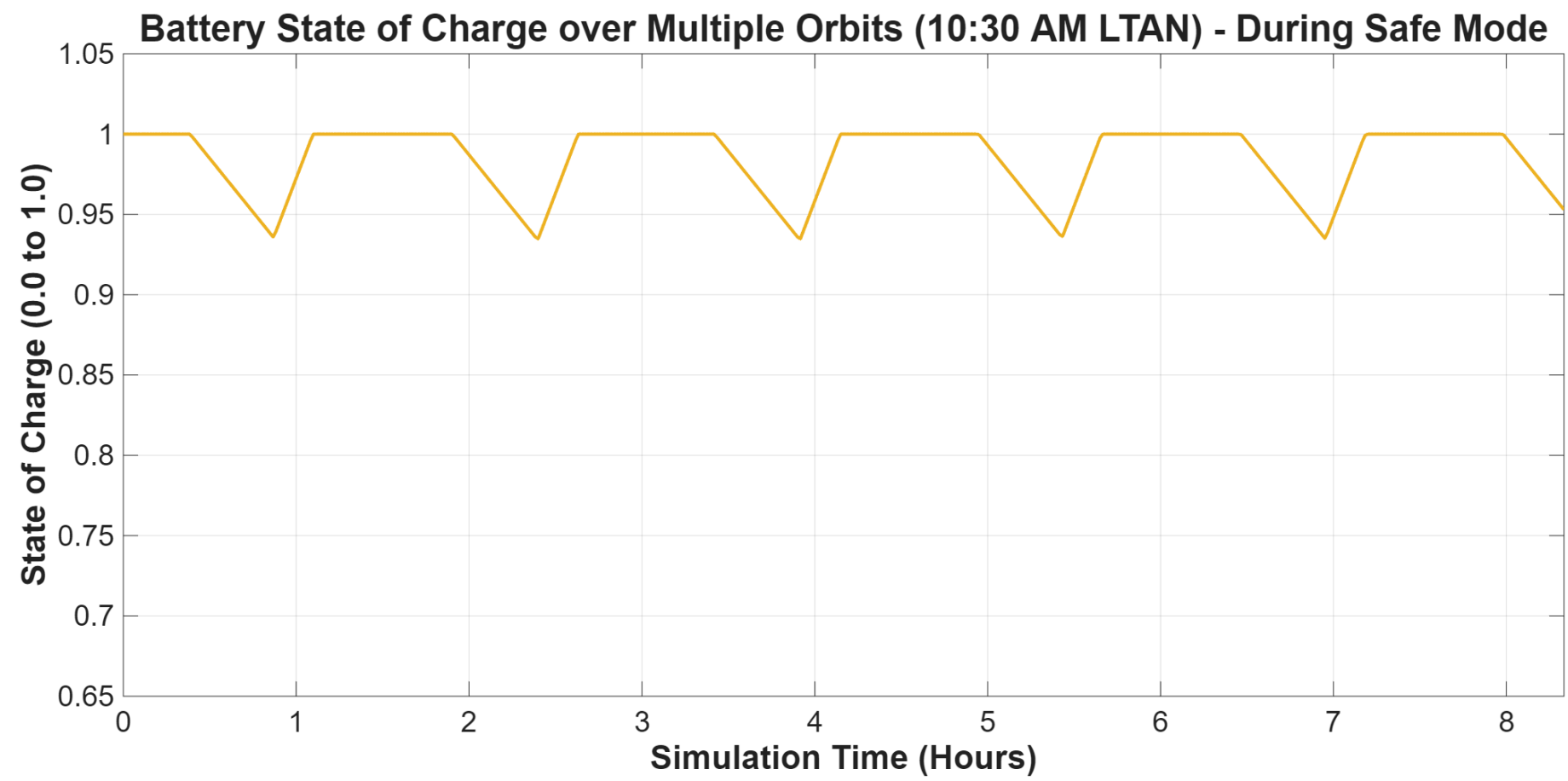


Figure B3: Battery in Extreme Conditions With Safe Mode

Appendix C: DEMS Verification Data

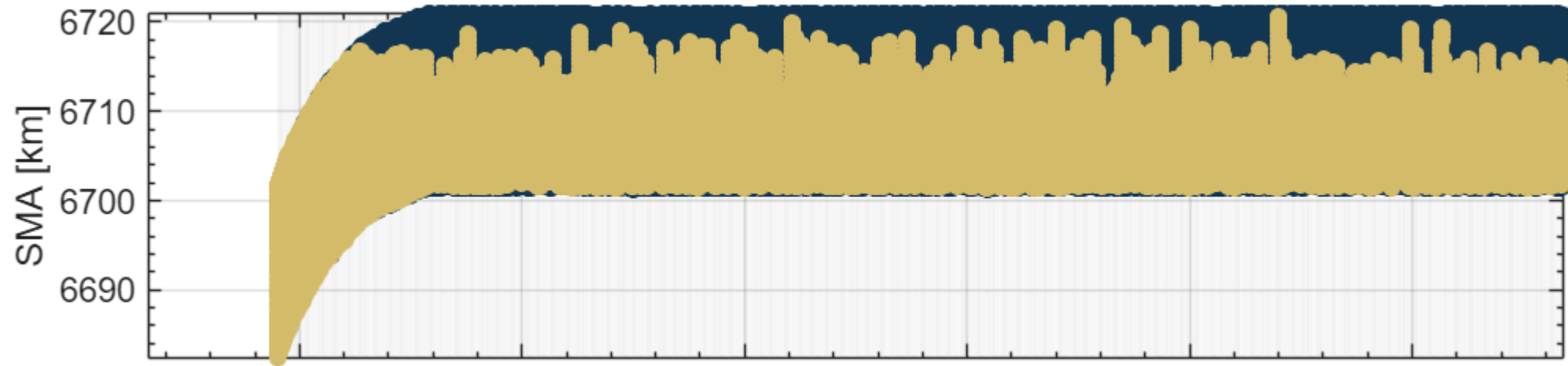


Figure C1: Semi-Major Axis across orbit

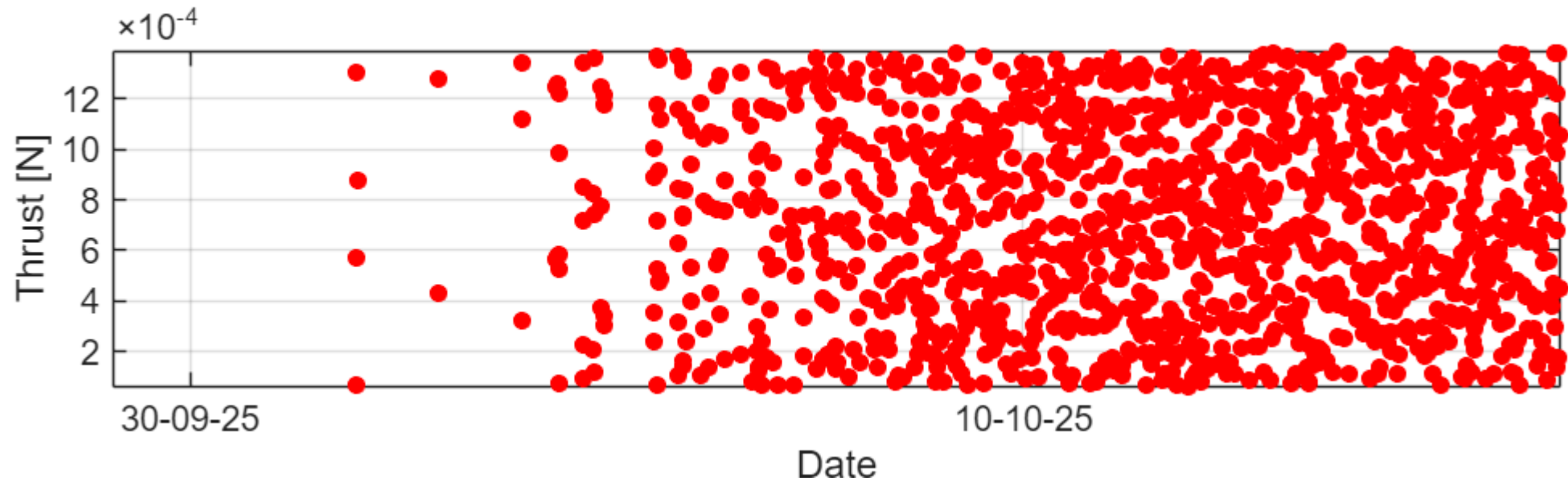


Figure C2: Thrust Action Across Simulation

Appendix D: Volume Study

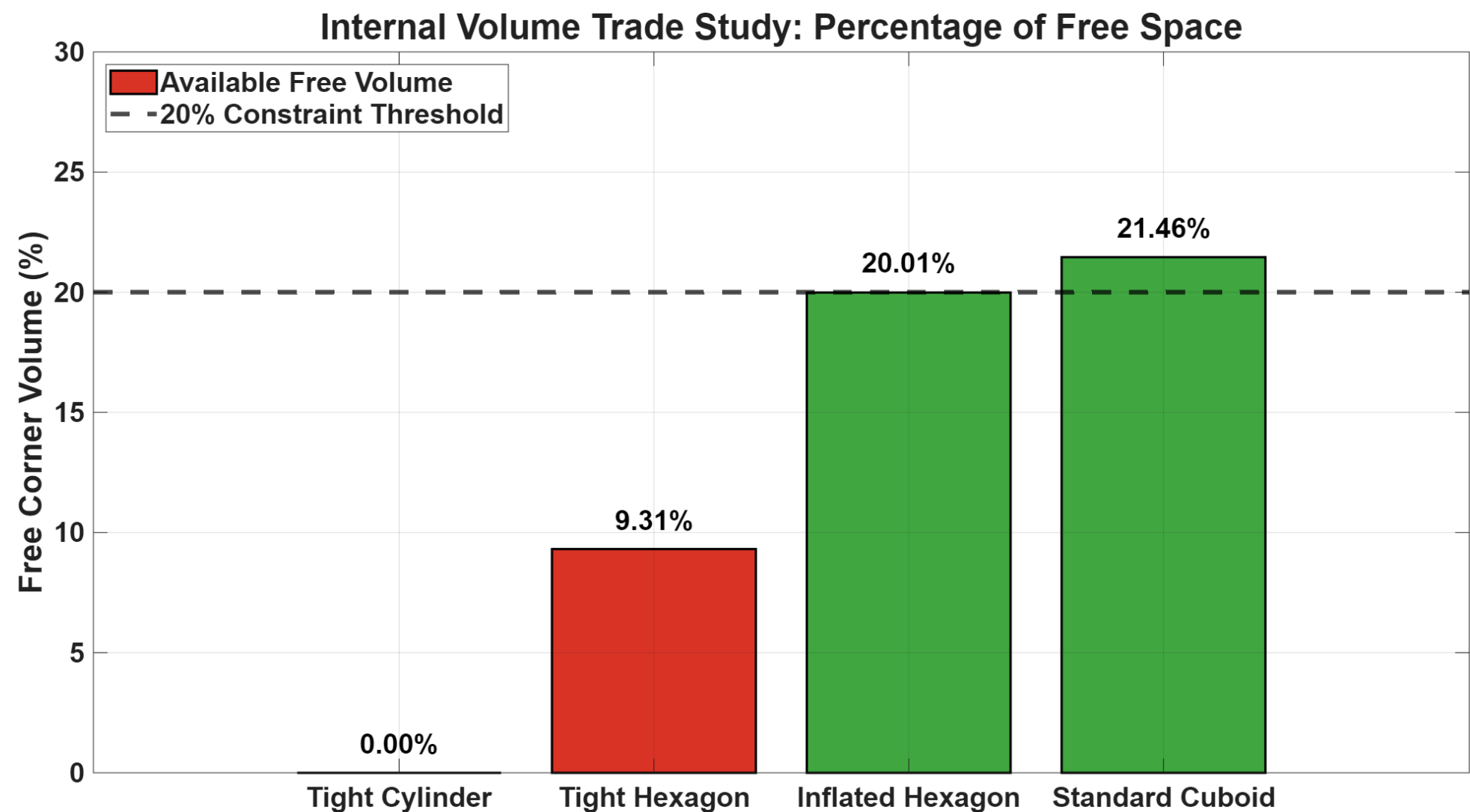


Figure D1: Volume across configurations